Part A: Literature Exploration and Comparison

Group Number: 153

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Domain: Medical Image Diagnosis

Criteria	**PAPER 1**	**PAPER 2**	**PAPER 3**
Title of the paper	A review of convolutional neural network based methods for medical image classification	Developments in Deep Learning Artificial Neural Network for Medical Image Analysis	3D TransUNet: Advancing Medical Image Segmentation through Vision Transformers
Authors	Chao Chen,Nor Ashidi Mat Isa,Xin Liu	Olamilekan Shobayo, Reza Saatchi	Jieneng Chen, Jieru Mei, Xianhang Li, Yongyi Lu, Qihang Yu, Qingyue Wei, Xiangde Luo, Yutong Xie, Ehsan Adeli, Yan Wang, Matthew Lungren, Lei Xing, Le Lu, Alan Yuille, Yuyin Zhou
Year of publication	2025	2025	2023
Network used	Convolutional Neural Networks (CNNs)	Recurrent Neural Networks (RNNs), often in hybrid models with CNNs	3D TransUNet - Vision Transformer + U-Net hybrid architecture with Transformer encoder and decoder

Depth of the network	This implementation uses a custom CNN with 8 convolutional layers and 3 dense layers.	The depth can vary, RNNs may be combined with deep CNNs for feature extraction.	Built upon nnU-Net architecture with Transformer encoder (1-layer or 12-layer Vision Transformer) and Transformer decoder with iterative coarse-to-fine refinement. Three configurations: Encoder-only, Decoder-only, Encoder+Decoder.
How is the network helping the overall task?	Primarily for feature extraction and classification. CNNs are excellent at identifying spatial hierarchies of features in images, which is key for diagnostic classification.	Useful for analyzing sequential data in medical imaging, such as monitoring disease progression over time from a series of scans. They can be used for classification and segmentation.	The network performs 3D medical image segmentation by combining CNN's localization abilities with Transformer's global context modeling. Transformer encoder captures long range dependencies among organs, while Transformer decoder performs mask classification using learnable organ queries with cross attention refinement for precise segmentation.

Loss function used	It covers various loss functions. For classification tasks like ours, Binary Cross-Entropy is a standard choice.	The paper mentions that for classification tasks, loss functions like cross-entropy are common. For segmentation, the Dice coefficient might be used.	Encoder-only: Hybrid segmentation loss with pixel wise Cross-Entropy + Dice Loss. Decoder-only: Hungarian matching loss $L = \lambda_0(Lce + Ldice) + \lambda_1 Lcls$ combining pixel wise classification loss and binary mask loss. Deep supervision applied at each decoder stage.
Evaluation / Performance metric used	For classification, common metrics include Accuracy, Precision, Recall, and F1-Score.	The paper discusses metrics such as Accuracy, Area Under the ROC Curve (AUC), and the Dice Coefficient for segmentation.	Dice Similarity Coefficient (DSC), Hausdorff Distance (HD), Average Surface Distance for 3D segmentation evaluation. Clinical metrics include sensitivity and specificity for medical validation.

Name of Dataset used. If a public dataset, provide the URL.	The paper reviews methods applied to various medical imaging datasets. For our implementation, we've used the Chest X-Ray Images (Pneumonia) dataset. URL: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia	This paper covers a range of datasets. RNNs are often applied to datasets with a temporal component, like sequential MRI or CT scans.	Multiple 3D medical datasets: • Synapse multi-organ CT dataset: https://www.synapse.org/#!Synapse:syn3193805/wiki/217789 • BraTS2021 brain tumor dataset: http://braintumorsegmentation.org/ • Medical Segmentation Decathlon (MSD): http://medicaldecathlon.com/ • Large scale pancreatic mass dataset (2930 CT scans) • BraTS2023 brain metastases dataset

Conclusion	**Paper 1 (CNN)** provides a comprehensive overview of how CNNs are the go to architecture for image classification tasks. Their ability to learn spatial features directly from images makes them highly effective for diagnosing conditions from static images like X-rays.	**Paper 2 (RNN)** highlights that RNNs, often in combination with CNNs, are particularly powerful when applied to medical datasets involving sequential or temporal progression, such as MRI or CT scans taken over time. Their ability to capture temporal dependencies provides significant advantages for analyzing disease evolution. However, the observation is that for tasks involving static images, such as a single chest X-ray, the added complexity of RNNs does not translate into improved performance. This makes them less aligned with the requirements of our classification task.	**Paper 3 (3D TransUNet)** introduces an advanced Transformer based architecture specifically designed for 3D medical image segmentation. The paper presents a hybrid approach combining Vision Transformers with U-Net architecture, featuring both Transformer encoder and decoder components. The Transformer encoder excels in multi organ segmentation by capturing global context relationships, while the Transformer decoder proves more beneficial for small target segmentation like tumors through iterative coarse to fine attention refinement. While this represents state of the art technology for 3D medical segmentation, it is designed for pixel level segmentation tasks rather than our binary classification requirement.

Justification for Implementation Choice:

We have chosen to base our implementation on the methodologies reviewed in **Paper 1**.

Here's why:

Suitability for the Task: Our project is a binary classification problem, determining if a chest X-ray shows signs of pneumonia. CNNs are exceptionally well suited for this kind of image classification task, as extensively documented in the review.

Dataset Compatibility: The Chest X-Ray (Pneumonia) dataset consists of static images, which is the ideal input for a standard CNN architecture. The dataset does not have a sequential element that would necessitate an RNN, nor does it require the pixel level precision of a 3D segmentation model like the TransUNet.

Feasibility and Clarity: The principles of using CNNs for medical image classification are well understood and provide a clear and effective path for implementation. The provided code in Part B, which uses a CNN, aligns perfectly with the methods described in this review paper, making it the most logical and practical choice for our project.